Inverse Laplace Transform

* **Definition**: If $\mathcal{L}\{f(t)\} = F(s)$, then $f(t)$ is called the Inverse Laplace transform of $F(s)$ and we write symbolically $\{f(t)\} = \mathcal{L}^{-1}\{F(s)\}$ where $\mathcal{L}^{-1}$ is called the inverse Laplace transformation operator.

* **Note**: It can be shown that if two continuous functions have the same Laplace transform, then the two functions are identical (Lerch's theorem).

* **List of inverse transforms of some elementary functions**:

No.	$F(s)$	$f(t) = \mathcal{L}^{-1}\{F(s)\}$	No.	$F(s)$	$f(t) = \mathcal{L}^{-1}\{F(s)\}$
1.	$\frac{1}{s}$, $s>0$	1	11.	$\frac{s-a}{(s-a)^2+b^2}$	$e^{at}\cos bt$
2.	$\frac{1}{s^{n+1}}$, $n=0,1,2,\dots$ $s>0$	$\frac{t^n}{n!}$	12.	$\frac{1}{(s-a)^2+b^2}$	$\frac{1}{b}e^{at}\sin bt$
3.	$\frac{1}{s^{n+1}}$, $n\neq -1,-2,\dots$ $s>0$	$\frac{t^n}{\Gamma(n+1)}$	13.	$\frac{s-a}{(s-a)^2-b^2}$	$e^{at}\cosh bt$
4.	$\frac{1}{s-a}$, $s>a$	e^{at}	14.	$\frac{1}{(s-a)^2-b^2}$	$\frac{1}{b}e^{at}\sinh bt$
5.	$\frac{1}{s^2+a^2}$	$\frac{1}{a}\sin at$	15.	$\frac{s}{(s^2+a^2)^2}$	$\frac{1}{2a}t\sin at$
6.	$\frac{s}{s^2+a^2}$	$\cos at$	16.	$\frac{s^2-a^2}{(s^2+a^2)^2}$	$t\cos at$
7.	$\frac{1}{s^2-a^2}$, $s>\lvert a\rvert$	$\frac{1}{a}\sinh at$	17.	$\frac{F(s)}{s}$	$\int_0^t f(t)\,dt$
8.	$\frac{s}{s^2-a^2}$	$\cosh at$	18.	$sF(s)-f(0)$	$f'(t)$
9.	$\frac{s+b}{(s+b)^2+a^2}$	$e^{-bt}\cos at$	19.	$s^2F(s)-sf(0)-f'(0)$	$f''(t)$
10.	$\frac{a}{(s+b)^2+a^2}$	$e^{-bt}\sin at$	20	$-\frac{d}{ds}[F(s)]$	$-tf(t)$

* Linearity Property:

If $F_1(s)$ and $F_2(s)$ are the Laplace transforms of $f_1(t)$ and $f_2(t)$ respectively, then

$$\mathcal{L}^{-1}\left[c_1F_1(s)+c_2F_2(s)\right] = c_1\mathcal{L}^{-1}\left[F_1(s)\right] + c_2\mathcal{L}^{-1}\left[F_2(s)\right] = c_1f_1(t)+c_2f_2(t)$$

where c_1 and c_2 are any constants.

* Find the inverse Laplace transforms of the following.

1. $\frac{1}{s^3}$

Sol $\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s^{2+1}}\right\} = \frac{t^2}{2!}$

2. $\frac{1}{2s-5}$

Sol $\mathcal{L}^{-1}\left\{\frac{1}{2s-5}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{2(s-5/2)}\right\} = \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{s-5/2}\right\} = \frac{1}{2}e^{5t/2}$

3. $\frac{s+b}{s^2+a^2}$

Sol $\mathcal{L}^{-1}\left\{\frac{s+b}{s^2+a^2}\right\} = \mathcal{L}^{-1}\left\{\frac{s}{s^2+a^2}\right\} + \mathcal{L}^{-1}\left\{\frac{b}{s^2+a^2}\right\} = \cos at + \frac{b}{a}\sin at$

4. $\frac{1}{3s^2+16}$

Sol $\mathcal{L}^{-1}\left\{\frac{1}{3s^2+16}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{3(s^2+16/3)}\right\} = \frac{1}{3}\mathcal{L}^{-1}\left\{\frac{1}{s^2+(4/\sqrt{3})^2}\right\}$

$$= \frac{1}{3}\cdot\frac{\sqrt{3}}{4}\sin\frac{4}{\sqrt{3}}t = \frac{1}{4\sqrt{3}}\sin\frac{4}{\sqrt{3}}t.$$

5. $\frac{2s-5}{4s^2+25} + \frac{4s-9}{9-s^2}$

Sol $\mathcal{L}^{-1}\left\{\frac{2s-5}{4s^2+25} + \frac{4s-9}{9-s^2}\right\} = \frac{2}{4}\mathcal{L}^{-1}\left\{\frac{s-5/2}{s^2+25/4}\right\} - 4\mathcal{L}^{-1}\left\{\frac{s-9/4}{s^2-9}\right\}$

$$= \frac{2}{4}\left[\cos\frac{5}{2}t - \sin\frac{5}{2}t\right] - 4\left[\cosh 3t - \frac{3}{4}\sinh 3t\right].$$

* Shifting property for inverse Laplace transforms.

WKT, if $L\{f(t)\} = F(s)$, then $L\{e^{at} f(t)\} = F(s-a)$

From this result it follows that, if $L^{-1}\{F(s)\} = f(t)$, then

$$L^{-1}\{F(s-a)\} = e^{at} f(t) = e^{at} L^{-1}\{F(s)\}.$$

Examples:

1. Evaluate $L^{-1}\left\{\frac{3s+1}{(s+1)^4}\right\}$

Soln

$$L^{-1}\left\{\frac{3s+1}{(s+1)^4}\right\} = L^{-1}\left\{\frac{3s+3-2}{(s+1)^4}\right\}$$

$$= L^{-1}\left\{\frac{3(s+1)}{(s+1)^4}\right\} - 2L^{-1}\left\{\frac{1}{(s+1)^4}\right\}$$

$$= 3L^{-1}\left\{\frac{1}{(s+1)^3}\right\} - 2L^{-1}\left\{\frac{1}{(s+1)^4}\right\}$$

$$= 3 \cdot e^{-t} \cdot \frac{t^2}{2!} - 2 \cdot e^{-t} \cdot \frac{t^3}{3!} = e^{-t}\left(\frac{3t^2}{2} - \frac{t^3}{3}\right)$$

2. Evaluate $L^{-1}\left\{\frac{s+2}{s^2-2s+5}\right\}$

Soln

$$L^{-1}\left\{\frac{s+2}{s^2-2s+5}\right\} = L^{-1}\left\{\frac{s+2}{(s-1)^2+4}\right\} = L^{-1}\left\{\frac{(s-1)+3}{(s-1)^2+4}\right\}$$

$$= L^{-1}\left\{\frac{s-1}{(s-1)^2+4}\right\} + 3L^{-1}\left\{\frac{1}{(s-1)^2+4}\right\}$$

$$= e^{t}\cos 2t + \frac{3}{2}e^{t}\sin 2t$$

3. Evaluate $L^{-1}\left\{\frac{2s+1}{s^2+3s+1}\right\}$

Soln

$$L^{-1}\left\{\frac{2s+1}{s^2+3s+1}\right\} = 2L^{-1}\left\{\frac{s+1/2}{(s+3/2)^2+1-\frac{9}{4}}\right\} = 2L^{-1}\left\{\frac{(s+3/2)-1}{(s+3/2)^2-5/4}\right\}$$

$$= 2L^{-1}\left\{\frac{(s+3/2)}{(s+3/2)^2-5/4}\right\} - 2L^{-1}\left\{\frac{1}{(s+3/2)^2-5/4}\right\} = 2e^{-3/2\,t}\left[\cosh\frac{\sqrt{5}}{2}t - \frac{2}{\sqrt{5}}\sinh\frac{\sqrt{5}}{2}t\right]$$

4. Evaluate $\mathcal{L}^{-1}\left\{\frac{3s+2}{2s^2-4s+3}\right\}$

Soln

$$\frac{3s+2}{2s^2-4s+3} = \frac{1}{2}\left[\frac{3s+2}{s^2-2s+3/2}\right] = \frac{1}{2}\left[\frac{3s-3+3+2}{(s-1)^2+\left(\frac{1}{\sqrt{2}}\right)^2}\right]$$

$$\therefore \mathcal{L}^{-1}\left\{\frac{3s+2}{2s^2-4s+3}\right\} = \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{3(s-1)}{(s-1)^2+\left(\frac{1}{\sqrt{2}}\right)^2} + \frac{5}{(s-1)^2+\left(\frac{1}{\sqrt{2}}\right)^2}\right\}$$

$$= \frac{e^t}{2}\left\{3\cos\frac{t}{\sqrt{2}} + 5\sqrt{2}\sin\frac{t}{\sqrt{2}}\right\}$$

5. Use the method of partial fractions to find the time signals corresponding to the Laplace transform function

$$F(s) = \frac{4s+5}{(s+1)^2(s+2)}$$

Soln

$$\frac{4s+5}{(s+1)^2(s+2)} = \frac{A}{(s+1)^2} + \frac{B}{(s+1)} + \frac{C}{(s+2)}$$

Then $4s+5 = A(s+2) + B(s+1)(s+2) + C(s+1)^2$

For $s=-1$, $A=1$; For $s=-2$, $C=-3$

comparing the coefficients of s^2, we get $B+C=0 \Rightarrow B=3$

$$\therefore \frac{4s+5}{(s+1)^2(s+2)} = \frac{1}{(s+1)^2} + \frac{3}{(s+1)} - \frac{3}{(s+2)}$$

Hence, $\mathcal{L}^{-1}\left\{\frac{4s+5}{(s+1)^2(s+2)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2}\right\} + 3\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} - 3\mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\}$

$$= e^{-t}.t + 3e^{-t} - 3e^{-2t}$$

6. Evaluate the following

(i) $\mathcal{L}^{-1}\left\{\frac{2s^2+5s-4}{s^3+s^2-2s}\right\}$

Soln

$$\frac{2s^2+5s-4}{s^3+s^2-2s} = \frac{2s^2+5s-4}{s(s^2+s-2)} = \frac{2s^2+5s-4}{s(s+2)(s-1)} = \frac{A}{s} + \frac{B}{s-1} + \frac{C}{s+2}$$

$\Rightarrow 2s^2+5s-4 = A(s-1)(s+2) + Bs(s+2) + Cs(s-1)$

For $s=1$, $3 = 3B \Rightarrow B=1$

$s=-2$, $-6 = 6C \Rightarrow C=-1$

$s=0$, $-4 = -2A \Rightarrow A=2$

$$\therefore \mathcal{L}^{-1}\left\{\frac{2s^2+5s-4}{s^3+s^2-2s}\right\} = \mathcal{L}^{-1}\left\{\frac{2}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\}$$

$$= 2 + e^{t} - e^{-2t}$$

(ii) $\mathcal{L}^{-1}\left\{\frac{s^3}{s^4-a^4}\right\}$

Soln $s^4-a^4 = (s^2+a^2)(s^2-a^2) = (s^2+a^2)(s+a)(s-a)$

$$\therefore \frac{s^3}{s^4-a^4} = \frac{s^3}{(s^2+a^2)(s+a)(s-a)} = \frac{As+B}{(s^2+a^2)} + \frac{C}{(s+a)} + \frac{D}{(s-a)}$$

$$\Rightarrow s^3 = (As+B)(s^2-a^2) + C(s-a)(s^2+a^2) + D(s+a)(s^2+a^2).$$

For $s=a$, $a^3 = 4a^3D \Rightarrow D = 1/4$

$s=-a$, $-a^3 = -4a^3C \Rightarrow C = 1/4$

Comparing the constant terms, we get $0 = -Ba^2 - \cancel{Ca^3} + \cancel{Da^3}$
$\Rightarrow B=0.$

Comparing the coefficients of s^3, we get $1 = A+C+D = A+1/4+1/4$
$\Rightarrow A = 1/2.$

$$\therefore \mathcal{L}^{-1}\left\{\frac{s^3}{s^4-a^4}\right\} = \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{s}{s^2+a^2}\right\} + \frac{1}{4}\mathcal{L}^{-1}\left\{\frac{1}{s+a} + \frac{1}{s-a}\right\}$$

$$= \frac{1}{2}\cos at + \frac{1}{4}(e^{at}+e^{-at})$$

$$= \frac{1}{2}\left[\cos at + \cosh at\right]$$

(iii) $\mathcal{L}^{-1}\left\{\frac{s}{s^4+s^2+1}\right\}$ $\qquad s^4+s^2+1 = (s^2+1)^2 - s^2$

Soln

$$\frac{s}{s^4+s^2+1} = \frac{s}{(s^2+s+1)(s^2-s+1)} = \frac{1}{2}\left[\frac{2s}{(s^2+s+1)(s^2-s+1)}\right]$$

$$= \frac{1}{2}\left[\frac{(s^2+s+1)-(s^2-s+1)}{(s^2+s+1)(s^2-s+1)}\right] = \frac{1}{2}\left[\frac{1}{(s^2-s+1)} - \frac{1}{(s^2+s+1)}\right]$$

$$= \frac{1}{2}\left[\frac{1}{(s-1/2)^2+1-\frac{1}{4}} - \frac{1}{(s+1/2)^2+1-1/4}\right] = \frac{1}{2}\left[\frac{1}{(s-1/2)^2+\frac{3}{4}} - \frac{1}{(s+1/2)^2+3/4}\right]$$

$$\therefore \mathcal{L}^{-1}\left\{\frac{s}{s^4+s^2+1}\right\} = \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{(s-\frac{1}{2})^2+\frac{3}{4}} - \frac{1}{(s+\frac{1}{2})^2+\frac{3}{4}}\right\}$$

$$= \frac{1}{2}\left[e^{\frac{1}{2}t}\cdot\sqrt{\frac{4}{3}}\cdot\sin\sqrt{\frac{3}{4}}t - e^{-\frac{1}{2}t}\sqrt{\frac{4}{3}}\sin\sqrt{\frac{3}{4}}t\right]$$

$$= \frac{2}{\sqrt{3}}\sin\frac{\sqrt{3}}{2}t\left(\frac{e^{t/2}-e^{-t/2}}{2}\right)$$

$$= \frac{2}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right)\sinh\left(t/2\right)$$

(iv) $\mathcal{L}^{-1}\left\{\frac{5s+3}{(s-1)(s^2+2s+5)}\right\}$

Soln

$$\frac{5s+3}{(s-1)(s^2+2s+5)} = \frac{A}{(s-1)} + \frac{Bs+C}{(s^2+2s+5)}$$

$$\Rightarrow 5s+3 = A(s^2+2s+5) + (Bs+C)(s-1)$$

For $s=1$, $8 = 8A \Rightarrow A = 1$.

Comparing the coefficients of s^2, we get $0 = A+B \Rightarrow B=-1$.

For $s=0$, $3 = 5A + (-C) \Rightarrow C = 2$

$$\therefore \mathcal{L}^{-1}\left\{\frac{5s+3}{(s-1)(s^2+2s+5)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} + \mathcal{L}^{-1}\left\{\frac{-s+2}{s^2+2s+5}\right\}$$

$$= e^t + \mathcal{L}^{-1}\left\{\frac{-s+2}{(s+1)^2+4}\right\}$$

$$= e^t + \mathcal{L}^{-1}\left\{\frac{3-(s+1)}{(s+1)^2+2^2}\right\}$$

$$= e^t + \mathcal{L}^{-1}\left\{\frac{3}{(s+1)^2+2^2}\right\} + \mathcal{L}^{-1}\left\{\frac{-(s+1)}{(s+1)^2+2^2}\right\}$$

$$= e^t + \frac{3}{2}e^{-t}\sin 2t - e^{-t}\cos 2t$$

$$= e^t + e^{-t}\left(\frac{3}{2}\sin 2t - \cos 2t\right)$$

* Evaluation of $\mathcal{L}^{-1}\{e^{-as}F(s)\}$ (Heaviside Shift theorem)

We know that, if $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$

From this result, it follows that $\mathcal{L}^{-1}\{e^{-as}F(s)\} = f(t-a)u(t-a)$

Working procedure : In the given function observe the presence of e^{-as} first and identify the remaining part of the function to be called as $F(s)$. Taking the inverse of $F(s)$ we obtain $f(t)$. The required inverse of $e^{-as}F(s)$ is obtained by replacing t by $(t-a)$ in $f(t)$ to be multiplied by the unit step function $u(t-a)$.

Examples :

1. Evaluate $\mathcal{L}^{-1}\left\{\frac{e^{-5s}}{(s-2)^4}\right\}$

Soln $F(s) = \frac{1}{(s-2)^4}$, $a = 5$

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{(s-2)^4}\right\} = e^{2t}\mathcal{L}^{-1}\left\{\frac{1}{s^4}\right\} = e^{2t}\frac{t^3}{3!}$$

$$\mathcal{L}^{-1}\left\{\frac{e^{-5s}}{(s-2)^4}\right\} = f(t-a)u(t-a) = f(t-5)u(t-5)$$

$$= \frac{e^{2(t-5)}(t-5)^3}{6}u(t-5)$$

2. Evaluate $\mathcal{L}^{-1}\left\{\frac{e^{-\pi s}}{s^2+1} + \frac{se^{-2\pi s}}{s^2+4}\right\}$

Soln $\mathcal{L}^{-1}\left\{\frac{e^{-\pi s}}{s^2+1} + \frac{se^{-2\pi s}}{s^2+4}\right\} = f_1(t-\pi)u(t-\pi) + f_2(t-2\pi)u(t-2\pi)$

$$f_1(t) = \mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin t$$

$$f_2(t) = \mathcal{L}^{-1}\left\{\frac{s}{s^2+4}\right\} = \cos 2t$$

$$\therefore \mathcal{L}^{-1}\left\{\frac{e^{-\pi s}}{s^2+1} + \frac{se^{-2\pi s}}{s^2+4}\right\} = \sin(t-\pi)u(t-\pi) + \cos 2(t-2\pi)u(t-2\pi)$$

$$= -\sin t\, u(t-\pi) + \cos 2t\, u(t-2\pi)$$

3. $\mathcal{L}^{-1}\left\{\dfrac{(s+1)e^{-\pi s}}{s^2+s+1}\right\}$

Sol: Consider $F(s) = \dfrac{s+1}{s^2+s+1} = \dfrac{s+1}{(s+\frac{1}{2})^2+\frac{3}{4}} = \dfrac{(s+\frac{1}{2})+\frac{1}{2}}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2}$

$$\mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{(s+\frac{1}{2})}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2}\right\} + \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2}\right\}$$

$$= e^{-\frac{1}{2}t}\cos\left(\frac{\sqrt{3}}{2}t\right) + \frac{1}{2}\cdot e^{-\frac{1}{2}t}\cdot\frac{2}{\sqrt{3}}\cdot\sin\left(\frac{\sqrt{3}}{2}t\right)$$

$$= e^{-\frac{1}{2}t}\left[\cos\frac{\sqrt{3}}{2}t + \frac{1}{\sqrt{3}}\sin\frac{\sqrt{3}}{2}t\right]$$

By using the Heaviside shift property,

$$\mathcal{L}^{-1}\left\{\frac{(s+1)e^{-\pi s}}{s^2+s+1}\right\} = e^{-\frac{1}{2}(t-\pi)}\left\{\cos\frac{\sqrt{3}}{2}(t-\pi)\,H(t-\pi) + \frac{1}{\sqrt{3}}\sin\frac{\sqrt{3}}{2}(t-\pi)H(t-\pi)\right\}$$

* Evaluation of $\mathcal{L}^{-1}\left\{\dfrac{F(s)}{s}\right\}$ and $\mathcal{L}^{-1}\{F^n(s)\}$

We know that (i) $\mathcal{L}\{t^n f(t)\} = (-1)^n F^n(s)$

(ii) $\mathcal{L}\left\{\int_0^t f(t)\,dt\right\} = \dfrac{F(s)}{s}$, where $\mathcal{L}\{f(t)\} = F(s)$.

From these results, it follows that

(i) $\mathcal{L}^{-1}\left\{(-1)^n \dfrac{d^n}{ds^n}F(s)\right\} = t^n f(t)$

(ii) $\mathcal{L}^{-1}\left\{\dfrac{F(s)}{s}\right\} = \int_0^t f(t)\,dt$

In particular, (i) $\mathcal{L}^{-1}\left\{-\dfrac{d}{ds}F(s)\right\} = t f(t)$

(ii) $\mathcal{L}^{-1}\left\{\dfrac{d^2}{ds^2}F(s)\right\} = t^2 f(t)$

Note: $\mathcal{L}^{-1}\{F^n(s)\} = (-1)^n t^n f(t) = (-1)^n t^n \mathcal{L}^{-1}\{F(s)\}$ (or)

$$\mathcal{L}^{-1}\{F(s)\} = \frac{(-1)^n}{t^n}\mathcal{L}^{-1}\{F^n(s)\}$$

Examples :

1. Evaluate $\mathcal{L}^{-1}\left\{\log\left(\frac{s+a}{s+b}\right)\right\}$

Sol: Let $F(s) = \log\left(\frac{s+a}{s+b}\right) = \log(s+a) - \log(s+b)$

$$-\frac{d}{ds}F(s) = -\left[\frac{1}{s+a} - \frac{1}{s+b}\right] = \frac{1}{s+b} - \frac{1}{s+a}$$

$$\mathcal{L}^{-1}\left\{-\frac{d}{ds}F(s)\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s+b} - \frac{1}{s+a}\right\} = t\,f(t)$$

$$= e^{-bt} - e^{-at} = t\,f(t)$$

$$\Rightarrow f(t) = \frac{e^{-bt} - e^{-at}}{t}$$

2. Evaluate $\mathcal{L}^{-1}\left\{\tan^{-1}\frac{a}{s}\right\}$

Sol: Let $F(s) = \tan^{-1}\frac{a}{s}$

$$-\frac{d}{ds}F(s) = -\frac{1}{1+\frac{a^2}{s^2}}\cdot\left(\frac{-a}{s^2}\right) = \frac{a}{s^2+a^2}$$

$$\mathcal{L}^{-1}\left\{-\frac{d}{ds}F(s)\right\} = \mathcal{L}^{-1}\left\{\frac{a}{s^2+a^2}\right\}$$

$$\Rightarrow t\,f(t) = \sin at$$

$$\Rightarrow f(t) = \frac{\sin at}{t}$$

3. Find $\mathcal{L}^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\}$

Sol: WKT $\mathcal{L}^{-1}\left\{\frac{1}{(s^2+a^2)}\right\} = \frac{1}{a}\sin at$

$$\mathcal{L}^{-1}\left\{-\frac{d}{ds}\left(\frac{1}{s^2+a^2}\right)\right\} = t\cdot\frac{1}{a}\sin at$$

$$\therefore \mathcal{L}^{-1}\left\{\frac{2s}{(s^2+a^2)^2}\right\} = \frac{1}{a}t\sin at \Rightarrow \mathcal{L}^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\} = \frac{t\sin at}{2a}.$$

Examples:

1. Evaluate $\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+a^2)}\right\}$

Sol: Let $F(s) = \frac{1}{s^2+a^2}$. $\Rightarrow \mathcal{L}^{-1}\{F(s)\} = f(t) = \frac{\sin at}{a}$

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+a^2)}\right\} = \mathcal{L}^{-1}\left\{\frac{F(s)}{s}\right\} = \int_0^t f(t)\,dt = \int_0^t \frac{\sin at}{a}\,dt$$

$$= \frac{1}{a}\left[\frac{-\cos at}{a}\right]_0^t$$

$$= \frac{1}{a^2}(-\cos at + 1) = \frac{1}{a^2}(1-\cos at)$$

2. Evaluate $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s+a)^2}\right\}$

Sol: $\mathcal{L}^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}$

$$\mathcal{L}^{-1}\left\{-\frac{d}{ds}\left(\frac{1}{s+a}\right)\right\} = \mathcal{L}^{-1}\left\{\frac{1}{(s+a)^2}\right\} = te^{-at}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s+a)^2}\right\} = \int_0^t te^{-at}\,dt$$

$$= \left[t\cdot\frac{e^{-at}}{-a} - 1\cdot\frac{e^{-at}}{(-a)^2}\right]_0^t$$

$$= -\frac{te^{-at}}{a} - \frac{e^{-at}}{a^2} + \frac{1}{a^2}$$

$$= \frac{1}{a^2}\left[1 - e^{-at}(1+at)\right]$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2(s+a)^2}\right\} = \int_0^t \frac{1}{a^2}\left[1-e^{-at}(1+at)\right]dt$$

$$= \frac{1}{a^2}\left[t\Big|_0^t - (1+at)\frac{e^{-at}}{-a} + a\cdot\frac{e^{-at}}{(-a)^2}\right]_0^t$$

$$= \frac{1}{a^2}\left[t + \frac{(1+at)e^{-at}}{a} + \frac{1}{a}e^{-at} + \frac{e^{-at}}{-a} - \frac{1}{a}\right]$$

$$= \frac{1}{a^3}\left[at - 1 + e^{-at} + ate^{-at}\right] = \frac{1}{a^3}\left[at(1+e^{-at}) + (e^{-at}-1)\right]$$

3. Find $\mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s(s^2-1)}\right\}$

Sol: $\mathcal{L}^{-1}\left\{\frac{1}{s^2-1}\right\} = \sinh t$

$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2-1)}\right\} = \int_0^t \sinh t \, dt = [\cosh t]_0^t = \cosh t - 1$

$\therefore \mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s(s^2-1)}\right\} = [\cosh(t-2) - 1]\, u(t-2)$

* Convolution

The convolution of two functions $f(t)$ and $g(t)$ usually denoted by $f(t) * g(t)$ is defined by

$$f(t) * g(t) = \int_0^t f(u)\, g(t-u)\, du .$$

Property : convolution operation $*$ is commutative.

$$f(t) * g(t) = g(t) * f(t).$$

Convolution theorem :

$$\mathcal{L}\{f(t)\} . \mathcal{L}\{g(t)\} = \mathcal{L}\{f(t) * g(t)\}$$

$$\mathcal{L}\{f(t)\} . \mathcal{L}\{g(t)\} = \mathcal{L}\left\{ \int_0^t f(u) g(t-u) du \right\}$$

$$= \mathcal{L}\left\{ \int_0^t f(t-u) g(u)\, du \right\}$$

1. Verify the convolution theorem for the functions $f(t) = t$, $g(t) = \cos t$.

Soln $\mathcal{L}\{f(t)\} = \frac{1}{s^2}$, $\mathcal{L}\{g(t)\} = \frac{s}{s^2+1}$

$$\therefore \mathcal{L}\{f(t)\} . \mathcal{L}\{g(t)\} = \frac{s}{s^2(s^2+1)} = \frac{1}{s(s^2+1)} \longrightarrow (1)$$

$$f(t) * g(t) = \int_0^t f(u)\, g(t-u)\, du$$

$$= \int_0^t u \cos(t-u)\, du$$

$$= \left[u \frac{\sin(t-u)}{-1} - 1 . \frac{-\cos(t-u)}{(-1)^2} \right]_0^t$$

$$= 1 - \cos t$$

$$\mathcal{L}\{f(t) * g(t)\} = \frac{1}{s} - \frac{s}{s^2+1} = \frac{s^2+1-s^2}{s(s^2+1)} = \frac{1}{s(s^2+1)} \rightarrow (2)$$

From (1) & (2) , $\mathcal{L}\{f(t)\} . \mathcal{L}\{g(t)\} = \mathcal{L}\{f(t) * g(t)\}$.

2. $f(t) = \sin t$, $g(t) = e^{-t}$

Sol: $\mathcal{L}\{f(t)\} = \frac{1}{s^2+1}$ $\quad \mathcal{L}\{g(t)\} = \frac{1}{s+1}$

$$\mathcal{L}\{f(t)\} \cdot \mathcal{L}\{g(t)\} = \frac{1}{(s^2+1)(s+1)}$$

$$f(t) * g(t) = \int_0^t f(u)\, g(t-u)\, du$$

$$= \int_0^t \sin u \; e^{-(t-u)}\, du$$

$$= e^{-t} \int_0^t e^u \sin u \, du$$

$$= e^{-t} \left[\frac{e^u}{2} (\sin u - \cos u) \right]_0^t$$

$$= \frac{e^{-t}}{2} \left[e^t (\sin t - \cos t) + 1 \right]$$

$$= \frac{\sin t - \cos t}{2} + \frac{e^{-t}}{2}$$

$$\mathcal{L}\{f(t) * g(t)\} = \frac{1}{2} \mathcal{L}\{\sin t - \cos t + e^{-t}\}$$

$$= \frac{1}{2} \left[\frac{1}{s^2+1} - \frac{s}{s^2+1} + \frac{1}{s+1} \right]$$

$$= \frac{1}{2} \left[\frac{\cancel{s}+1 - s(\cancel{s}+1) + \cancel{s^2}+1}{(s^2+1)(s+1)} \right]$$

$$= \frac{1}{2} \left[\frac{2}{(s^2+1)(s+1)} \right] = \frac{1}{(s^2+1)(s+1)}$$

3. S.T $\mathcal{L}\left\{ \int_0^t (t-u)\, e^{u-t} \cos u \, du \right\} = \frac{s}{(s+1)^2 (s^2+1)}$

Sol: $\int_0^t (t-u)\, e^{u-t} \cos u \, du = \int_0^t f(t-u)\, g(u) = f(t) * g(t)$

$f(t) = t e^{-t}$, $g(t) = \cos t$.

By C.T , we get $\mathcal{L}\left\{ \int_0^t (t-u)\, e^{u-t} \cos u \, du \right\} = \mathcal{L}\{f(t) * g(t)\}$

$$= \mathcal{L}\{f(t)\} \cdot \mathcal{L}\{g(t)\} = \mathcal{L}\{t e^{-t}\} \cdot \mathcal{L}\{\cos t\}$$

$$= \frac{1}{(s+1)^2} \cdot \frac{s}{s^2+1}$$

* Evaluation of Inverse Laplace Transform by using the Convolution theorem

If $\mathcal{L}^{-1}\{F(s)\} = f(t)$ and $\mathcal{L}^{-1}\{G(s)\} = g(t)$, then

$$\mathcal{L}^{-1}\{F(s)G(s)\} = \int_0^t f(u)g(t-u)\,du = f(t) * g(t).$$

Working rule:

S1. The given function is expressed as the product of two functions say $F(s)$ and $G(s)$.

S2. Find $\mathcal{L}^{-1}\{F(s)\} = f(t)$ and $\mathcal{L}^{-1}\{G(s)\} = g(t)$

S3. Apply the convolution theorem.

$$\mathcal{L}^{-1}\{F(s)G(s)\} = f(t) * g(t) = \int_0^t f(t-u)g(u)\,du = \int_0^t f(u)g(t-u)\,du$$

S4. Evaluate the convolution integral to obtain the required inverse.

Using convolution theorem, obtain the inverse Laplace transforms of the following functions.

(i) $\dfrac{1}{(s+a)(s+b)}$

Sol: Let $F(s) = \dfrac{1}{s+a}$, $G(s) = \dfrac{1}{s+b}$

Then $\mathcal{L}^{-1}\{F(s)\} = f(t) = e^{-at}$

$\mathcal{L}^{-1}\{G(s)\} = g(t) = e^{-bt}$

By convolution theorem,

$$L^{-1}\left\{\frac{1}{(s+a)(s+b)}\right\} = \int_0^t e^{-a(t-u)} e^{-bu}\, du$$

$$= e^{-at}\int_0^t e^{(a-b)u} du = e^{-at}\left[\frac{e^{(a-b)u}}{a-b}\right]_0^t$$

$$= \frac{e^{-at}}{a-b}\left[e^{(a-b)t}-1\right] = \frac{e^{-bt}-e^{-at}}{a-b}$$

(ii) $\dfrac{s}{(s^2+a^2)^2}$

Soln Let $F(s) = \dfrac{1}{s^2+a^2}$, $G(s) = \dfrac{s}{s^2+a^2}$

Then, $L^{-1}\{F(s)\} = f(t) = \dfrac{\sin at}{a}$; $L^{-1}\{G(s)\} = g(t) = \cos at$

By convolution theorem,

$$L^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\} = \int_0^t \frac{1}{a}\sin a(t-u)\cos au\, du$$

$$= \frac{1}{a}\int_0^t \frac{\sin at + \sin(at-2au)}{2}\, du$$

$$= \frac{1}{2a}\left[\sin at \,.u - \frac{\cos(at-2au)}{-2a}\right]_0^t$$

$$= \frac{1}{2a}\left[t\sin at + \frac{\cos at}{2a} - \frac{\cos at}{2a}\right] = \frac{t\sin at}{2a}.$$

(iii) $\dfrac{s}{(s-1)(s^2+1)}$

Soln Let $F(s) = \dfrac{1}{s-1}$, $G(s) = \dfrac{s}{s^2+1}$

Then $L^{-1}\{F(s)\} = f(t) = e^t$; $L^{-1}\{G(s)\} = g(t) = \cos t$

By convolution theorem,

$$L^{-1}\left\{\frac{1}{(s-1)(s^2+1)}\right\} = \int_0^t e^{t-u}\cos u\, du = e^t\int_0^t e^{-u}\cos u\, du$$

$$= e^t\left[e^{-u}\sin u\Big|_0^t + \int_0^t e^{-u}\sin u\, du\right]$$

$$= e^{t}\left[e^{-t}\sin t - e^{-u}\cos u\Big|_0^t - \int_0^t e^{-u}\cos u\,du\right]$$

$$= \frac{e^{t}}{2}\left[e^{-t}\sin t - e^{-t}\cos t + 1\right]$$

$$= \frac{1}{2}\left[e^{t} + \sin t - \cos t\right]$$

(iv) $\dfrac{s+2}{(s^2+4s+5)^2}$

Sol: Let $F(s) = \dfrac{s+2}{s^2+4s+5}$; $G(s) = \dfrac{1}{s^2+4s+5}$

Then $L^{-1}\{F(s)\} = L^{-1}\left\{\dfrac{s+2}{(s+2)^2+1}\right\} = e^{-2t}L^{-1}\left\{\dfrac{s}{s^2+1}\right\} = e^{-2t}\cos t = f(t)$

$L^{-1}\{G(s)\} = L^{-1}\left\{\dfrac{1}{(s+2)^2+1}\right\} = e^{-2t}L^{-1}\left\{\dfrac{1}{s^2+1}\right\} = e^{-2t}\sin t = g(t)$

By convolution theorem,

$$L^{-1}\left\{\frac{s+2}{(s^2+4s+5)^2}\right\} = \int_0^t e^{-2u}\cos u\; e^{-2(t-u)}\sin(t-u)\,du$$

$$= e^{-2t}\int_0^t \sin(t-u)\cos u\,du$$

$$= \frac{e^{-2t}}{2}\int_0^t [\sin(t-u+u) + \sin(t-u-u)]\,du$$

$$= \frac{e^{-2t}}{2}\int_0^t [\sin t + \sin(t-2u)]\,du$$

$$= \frac{e^{-2t}}{2}\left[\sin t\,.\,u - \frac{\cos(t-2u)}{-2}\right]_0^t$$

$$= \frac{e^{-2t}}{2}\left[t\sin t + \frac{\cos t}{2} - \frac{\cos t}{2}\right]$$

$$= \frac{e^{-2t}}{2}\,.\,t\sin t$$

* Laplace Transform Method for Differential Equations

As noted earlier Laplace transform technique is employed to solve initial-value problems. The solution of such a problem is obtained by using the Laplace Transform of the derivatives of function and then the inverse Laplace Transform.

The following are the expressions for the derivatives derived earlier.

$$\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0)$$
$$\mathcal{L}\{f''(t)\} = s^2\mathcal{L}\{f(t)\} - sf(0) - f'(0)$$
$$\mathcal{L}\{f'''(t)\} = s^3\mathcal{L}\{f(t)\} - s^2f(0) - sf'(0) - f''(0)$$

Examples :

1. Solve $y'' + 2y' - 3y = \sin t$, $y(0) = y'(0) = 0$.

Sol Taking Laplace transform of the given equation, we get

$$\mathcal{L}\{y'' + 2y' - 3y\} = \mathcal{L}\{\sin t\}$$

$$[s^2\mathcal{L}\{y(t)\} - sy(0) - y'(0)] + 2[s\mathcal{L}\{y(t)\} - y(0)] - 3\mathcal{L}\{y(t)\} = \frac{1}{s^2+1}$$

Using the given conditions, we get

$$s^2\mathcal{L}\{y(t)\} + 2s\mathcal{L}\{y(t)\} - 3\mathcal{L}\{y(t)\} = \frac{1}{s^2+1}$$

$$(s^2+2s-3)\mathcal{L}\{y(t)\} = \frac{1}{s^2+1}$$

$$(s-1)(s+3)\mathcal{L}\{y(t)\} = \frac{1}{s^2+1}$$

$$\mathcal{L}\{y(t)\} = \frac{1}{(s-1)(s+3)(s^2+1)}$$

$$y(t) = \mathcal{L}^{-1}\left\{\frac{1}{(s-1)(s+3)(s^2+1)}\right\}$$

Consider $\dfrac{1}{(s-1)(s+3)(s^2+1)} = \dfrac{A}{s-1} + \dfrac{B}{s+3} + \dfrac{Cs+D}{s^2+1}$

$$\Rightarrow \quad 1 = (s^2+1)(s+3)A + (s^2+1)(s-1)B + (Cs+D)(s-1)(s+3)$$

Put $s=1$. We get $1 = 8A \Rightarrow A = 1/8$

$s=-3 \Rightarrow 1 = -40B \Rightarrow B = \frac{-1}{40}$

Comparing coefficients of s^3, we get

$0 = A+B+C \Rightarrow \frac{1}{8} - \frac{1}{40} + C = 0 \Rightarrow C = \frac{-1}{10}$

Put $s=0$, we get $1 = 3A - B - 3D$

$\Rightarrow 1 = \frac{3}{8} + \frac{1}{40} - 3D \Rightarrow D = \frac{-1}{5}$

$$\therefore y(t) = \mathcal{L}^{-1}\left\{ \frac{1}{8}\cdot\frac{1}{s-1} - \frac{1}{40}\cdot\frac{1}{s+3} - \frac{\frac{s}{10}+\frac{1}{5}}{s^2+1} \right\}$$

$$= \frac{1}{8}e^{t} - \frac{1}{40}e^{-3t} - \frac{1}{10}\cos t - \frac{1}{5}\sin t$$

2. Employ Laplace Transform method to solve the integral equation

$$f(t) = 1 + \int_0^t f(u)\sin(t-u)\,du$$

Soln Taking Laplace transform of the given equation, we get

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{1\} + \mathcal{L}\left\{\int_0^t f(u)\sin(t-u)\,du\right\}$$

By using convolution theorem, we get

$$\mathcal{L}\{f(t)\} = \frac{1}{s} + \mathcal{L}\{f(t)\}\cdot\mathcal{L}\{\sin t\}$$

$$\mathcal{L}\{f(t)\}\left(1 - \frac{1}{s^2+1}\right) = \frac{1}{s}$$

$$\mathcal{L}\{f(t)\} = \frac{s^2+1}{s^3}$$

$$\Rightarrow f(t) = \mathcal{L}^{-1}\left\{\frac{s^2+1}{s^3}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{1}{s} + \frac{1}{s^3}\right\}$$

$$f(t) = 1 + \frac{t^2}{2}$$

3. A particle is moving along a path satisfying the equation $\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 25x = 0$ where x denotes the displacement of the particle at time t. If the initial position of the particle is at $x = 20$ and the initial speed is 10, find the displacement of the particle at any time t using Laplace transforms.

Soln Given equation may be rewritten as

$$x''(t) + 6x'(t) + 25x(t) = 0$$

Here, the initial conditions are $x(0) = 20$; $x'(0) = 10$.

Taking the Laplace transform of the given equation, we get

$$\mathcal{L}\{x''(t) + 6x'(t) + 25x(t)\} = \mathcal{L}\{0\}$$

$$[s^2\mathcal{L}\{x(t)\} - sx(0) - x'(0)] + 6[s\mathcal{L}\{x(t)\} - x(0)] + 25\mathcal{L}\{x(t)\} = 0$$

Using the given conditions, this becomes,

$$(s^2 + 6s + 25)\mathcal{L}\{x(t)\} - 20s - 10 - 120 = 0$$

$$\Rightarrow \quad \mathcal{L}\{x(t)\} = \frac{20s + 130}{s^2 + 6s + 25}$$

$$\Rightarrow \quad x(t) = \mathcal{L}^{-1}\left\{\frac{20s + 130}{(s+3)^2 + 16}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{20(s+3) + 70}{(s+3)^2 + 16}\right\}$$

$$= 20\,\mathcal{L}^{-1}\left\{\frac{s+3}{(s+3)^2 + 16}\right\} + 70\,\mathcal{L}^{-1}\left\{\frac{1}{(s+3)^2 + 16}\right\}$$

$$= 20\,e^{-3t}\cos 4t + \frac{70}{4}\cdot e^{-3t}\sin 4t$$

$$x(t) = 20e^{-3t}\cos 4t + \frac{35}{2}e^{-3t}\sin 4t$$

4. A Voltage Ee^{-at} is applied at $t=0$ to a circuit of inductance L and resistance R. Show that the current at any time t is $\frac{E}{R-aL}\left[e^{-at}-e^{-\frac{Rt}{L}}\right]$.

Sol: The circuit is an LR circuit.
The differential equation with respect to the circuit is

$$L\frac{di}{dt}+Ri=E(t)$$

Here, L denotes the inductance, i denotes current at any time t and $E(t)$ denotes the EMF.
It is given that $E(t)=Ee^{-at}$.
Thus, we have

$$L\frac{di}{dt}+Ri=Ee^{-at} \quad (\text{or}) \quad Li'(t)+Ri(t)=Ee^{-at}$$

Taking Laplace transform on both sides, we get

$$\mathcal{L}\{Li'(t)+Ri(t)\}=\mathcal{L}\{Ee^{-at}\}$$

$$L\left[s\mathcal{L}\{i(t)\}-i(0)\right]+R\left[\mathcal{L}\{i(t)\}\right]=E\mathcal{L}\{e^{-at}\}$$

Since $i(0)=0$, we get

$$[sL+R]\mathcal{L}\{i(t)\}=\frac{E}{s+a}$$

$$\mathcal{L}\{i(t)\}=\frac{E}{(s+a)(sL+R)}$$

$$i(t)=\mathcal{L}^{-1}\left\{\frac{E}{(s+a)(sL+R)}\right\}=\frac{E}{L}\mathcal{L}^{-1}\left\{\frac{1}{(s+a)\left(s+\frac{R}{L}\right)}\right\}$$

Consider $\frac{1}{(s+a)(s+R/L)}=\frac{A}{(s+a)}+\frac{B}{(s+R/L)}$

$$\Rightarrow 1=A(s+R/L)+B(s+a)$$

Put $s=-a$. We get $1=A\left(-a+\frac{R}{L}\right)\Rightarrow A=\frac{L}{R-aL}$

Put $s=-R/L$. We get $1=B\left(-\frac{R}{L}+a\right)\Rightarrow B=\frac{-L}{R-aL}$

Thus $i(t)=\frac{E}{L}\mathcal{L}^{-1}\left\{\frac{L}{R-aL}\cdot\frac{1}{s+a}-\frac{L}{R-aL}\cdot\frac{1}{s+R/L}\right\}$

$$i(t) = \frac{E}{R-aL} \cdot \mathcal{L}^{-1}\left\{ \frac{1}{s+a} - \frac{1}{s+R/L} \right\}$$

$$= \frac{E}{R-aL}\left[e^{-at} - e^{-R/L\, t}\right]$$

5. Mass spring damper system can be modeled using Newton's and Hooke's law. The differential equation representing this system is $\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 25x = 4\sin\omega t$ with initial conditions $x(0) = x'(0) = 0$. Solve for $x(t)$. Take $\omega = 2$

Soln Given equation may be rewritten as

$$x''(t) + 6x'(t) + 25x(t) = 4\sin\omega t$$

Here the initial conditions are $x(0) = 0$, $x'(0) = 0$.

Taking the Laplace transform of the equation, we get

$$s^2\mathcal{L}\{x(t)\} - sx(0) - x'(0) + 6[s\mathcal{L}\{x(t)\} - x(0)] + 25\mathcal{L}\{x(t)\} = \mathcal{L}\{4\sin\omega t\}$$

$$\Rightarrow \mathcal{L}\{x(t)\}(s^2+6s+25) = \frac{4\omega}{s^2+\omega^2}$$

$$\therefore \mathcal{L}\{x(t)\} = \frac{4\omega}{(s^2+\omega^2)(s^2+6s+25)}$$

consider $\frac{4\omega}{(s^2+\omega^2)(s^2+6s+25)} = \frac{As+B}{s^2+\omega^2} + \frac{Cs+D}{(s^2+6s+25)}$

$$\Rightarrow \frac{8}{(s^2+4)(s^2+6s+25)} = \frac{As+B}{s^2+4} + \frac{Cs+D}{s^2+6s+25}$$

$$\Rightarrow 8 = (As+B)(s^2+6s+25) + (Cs+D)(s^2+4)$$

Put $s=0$, we get $25B + 4D = 8 \longrightarrow (1)$

Coeff. of s^3 $\quad A + C = 0 \longrightarrow (2)$

Coeff. of s^2 $\quad 6A + B + D = 0 \longrightarrow (3)$

Coeff of s $\quad 25A + 6B + 4C = 0. \longrightarrow (4)$

Solving (1) & (3), we get $24A - 21B = -8 \longrightarrow (5)$

(2) in (4) gives $21A + 6B = 0 \longrightarrow (6)$

Solving (5) & (6), we get

$$A = \frac{-16}{195}, \quad B = \frac{56}{195}, \quad C = \frac{16}{195}, \quad D = \frac{8}{39} = \frac{40}{195}$$

$$\Rightarrow x(t) = \mathcal{L}^{-1}\left\{ \frac{\frac{-16}{195}s + \frac{56}{195}}{s^2+4} + \frac{\frac{16}{195}s + \frac{40}{195}}{(s+3)^2+16} \right\}$$

$$= \frac{1}{195}\mathcal{L}^{-1}\left\{ \frac{-16s}{s^2+4} + \frac{28 \times 2}{s^2+4} \right\} + \frac{1}{195}\mathcal{L}^{-1}\left\{ \frac{16s}{(s+3)^2+16} + \frac{40}{(s+3)^2+16} \right\}$$

$$= \frac{1}{195}\left\{ -16\cos 2t + 28\sin 2t \right\} + \frac{1}{195}\left\{ e^{-3t}(16\cos 4t + 10\sin 4t) \right\}$$

$$x(t) = \frac{4}{195}\left\{ 7\sin 2t - 4\cos 2t \right\} + \frac{2}{195}\left\{ e^{-3t}(8\cos 4t + 5\sin 4t) \right\}$$

6. Solve $(D^2+9)y = 18t$, $y(0) = 0$, $y(\pi/2) = 1$.

Soln Given $y''(t) + 9y(t) = 18t$

Taking Laplace transform on both sides, we get

$$\mathcal{L}\{y''(t) + 9y(t)\} = \mathcal{L}\{18t\}$$

$$\left[s^2\mathcal{L}\{y(t)\} + s\,y(0) - y'(0)\right] + 9\mathcal{L}\{y(t)\} = 18 \cdot \frac{1}{s^2}$$

Let $y'(0) = A$

$$\Rightarrow (s^2+9)\mathcal{L}\{y(t)\} - A = \frac{18}{s^2}$$

$$(s^2+9)\mathcal{L}\{y(t)\} = \frac{18}{s^2} + A$$

$$\mathcal{L}\{y(t)\} = Y(s) = \frac{18}{s^2(s^2+9)} + \frac{A}{(s^2+9)}$$

$$= \frac{18}{9}\left(\frac{1}{s^2} - \frac{1}{s^2+9} \right) + \frac{A}{s^2+9}$$

$$\therefore y(t) = \mathcal{L}^{-1}\left\{ \frac{2}{s^2} + \frac{A-2}{s^2+9} \right\}$$

$$y(t) = 2t + \frac{A-2}{3}\sin 3t$$

Substituting $y(\pi/2) = 1$, we get

$$1 = 2 \cdot \frac{\pi}{2} + \frac{A-2}{3} \sin \frac{3\pi}{2}$$

$$1 = \pi - \frac{A-2}{3}$$

$$3 = 3\pi - A + 2$$

$$A = 3\pi - 1$$

Hence, $y(t) = 2t + \frac{3\pi - 3}{3} \sin 3t$

$$\Rightarrow \quad y(t) = 2t + (\pi - 1) \sin 3t$$